

Claas Beger

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Education

Cornell University <i>MEng in Computer Science</i>	<i>Aug 2024 – May 2025</i> GPA: 4.21/4.3
Hong Kong University of Science and Technology <i>Exchange - Computer Science</i>	<i>Jan 2022 – May 2022</i> GPA: 4.1/4.3
Technical University of Munich <i>BS in Information Systems</i>	<i>Oct 2019 – Mar 2024</i> GPA: 3.7/4.0

Experience

Research Scientist <i>Stealth Startup</i>	<i>New York, New York</i> <i>July 2026 – Present</i>
- Developing label-free distillation methods for training models without human-annotated supervision. - Building evaluations that characterize jagged intelligence, i.e. uneven capability profiles across tasks.	
Graduate Fellow <i>Santa Fe Institute</i>	<i>Santa Fe, New Mexico</i> <i>June 2025 – June 2026</i>
- Researching multimodal abstract reasoning in AI models under Professor Melanie Mitchell. - Designing evaluation benchmarks and analysis pipelines for scientific reasoning with Sandia National Laboratories.	
Graduate Teaching and Research Specialist <i>Cornell University</i>	<i>Ithaca, New York</i> <i>Jan 2025 – May 2025</i>
- Co-designed assignments, projects, and exams for <i>Introduction to Machine Learning</i>. - Graded and provided feedback on students' work; held office hours for Q&A.	
Responsible AI Engineering Intern <i>PwC</i>	<i>Munich, Bavaria</i> <i>June 2022 – Jan 2023</i>
- Integrated SHAP and LIME into an explainable AI interface for TensorFlow, PyTorch, and sklearn models. - Co-authored and presented a white paper at Bitkom Big Data & AI Summit Berlin 2022.	
Financial Technologies SWE Intern <i>Deutsche Bank</i>	<i>Frankfurt, Hesse</i> <i>Oct 2021 – Jan 2022</i>
- Developed microservices in Python and Java (Spring Boot) on Red Hat OpenShift. - Managed database migration and analysis workflows using Flywaydb, JDBC, JPA, and PostgreSQL. - Collaborated with teams to integrate services into trading and risk management systems.	
Research Assistant <i>Institute of Computational Biology Munich</i>	<i>Munich, Bavaria</i> <i>Sep 2020 – Jun 2021</i>
- Built neural networks for classification and clustering of 20+ million single-cell samples using TensorFlow and Keras. - Developed autoencoders to generate synthetic diabetic cell samples to augment datasets. - Processed and visualized biological data in collaboration with computational biologists.	
Teaching Assistant <i>TU Munich; Chairs of Information Systems and Applied Software Engineering</i>	<i>Munich, Bavaria</i> <i>Oct 2020 – Sep 2021</i>
- Led weekly tutorials on software engineering, design patterns, model-based engineering, and information systems. - Developed exam questions, assisted with grading and lecture preparation, and held office hours for student support.	

Technologies

Technologies: Python, C++, Java, Swift, vLLM, PyTorch, TensorFlow, peft, DeepSpeed, Pandas, SpringBoot
Skills: Machine Learning, NLP, Computer Vision, Software Engineering, Neuromorphic Algorithms

Publications

To Appear

PotARCI: Multi-Dimensional Evaluation of Skill Acquisition in Abstract Reasoning Tasks. *Claas Beger*, Ryan Yi, Melanie Mitchell. To appear in Advances in Neural Information Processing Systems (NeurIPS 2026). [arXiv](#) 

Distinguishing Performance From Competence in Evaluations of Humanlike Abstract Reasoning. *Claas Beger*, Ryan Yi, Shuhao Fu, Arseny Moskvichev, Sarah W. Tsai, Sivasankaran Rajamanickam, Melanie Mitchell. To appear in Advances in Neural Information Processing Systems (NeurIPS 2026). [arXiv](#) 

Illusions of Abstraction in Contemporary AI. Melanie Mitchell, Jacob G. Foster, *Claas Beger*, Ryan Yi, Shuhao Fu, Kaleda K. Denton, Alessandro Palmarini. To appear in Proceedings of the National Academy of Sciences (PNAS).

Published

π -CoT: Prolog-Initialized Chain-of-Thought Prompting for Multi-Hop Question-Answering. Chao Wan, Albert Gong, Mihir Mishra, Carl-Leander Henneking, *Claas Beger*, Kilian Q. Weinberger. Transactions on Machine Learning Research (TMLR), 2026. [arXiv](#) 

OmniCode: A Benchmark for Evaluating Software Development Agents. Atharv Sonwane*, Eng-Shen Tu*, Wei-Chung Lu*, *Claas Beger**, ..., Kevin Ellis, Saikat Dutta. Findings of the Association for Computational Linguistics: ACL 2026. [Paper](#)  [arXiv](#) 

Bongards at the Boundary of Perception and Reasoning: Programs or Language? Cassidy Langenfeld, *Claas Beger*, Gloria Geng, Wasu Top Piriyaikulij, Keya Hu, Yewen Pu, Kevin Ellis. 48th Annual Meeting of the Cognitive Science Society (CogSci 2026). [arXiv](#) 

Cognitive-Science-Inspired Evaluation of Large Language Models. Shuhao Fu, *Claas Beger*, Ryan Yi, Melanie Mitchell, Michael C. Frank, Raphaël Millière, Robert Hawkins, Tomer D. Ullman. Symposium at the 48th Annual Meeting of the Cognitive Science Society (CogSci 2026). [Paper](#) 

Investigating Abstraction Capabilities of the o3 Model Using Textual and Visual Modalities. *Claas Beger*, Shuhao Fu, Ryan Yi, Arseny Moskvichev, Melanie Mitchell. NeurIPS 2025 Workshop on Multimodal Algorithmic Reasoning, Nov 2025. [Paper](#)  • **Spotlight Presentation**

Refining Inverse Constitutional AI for Dataset Validation Under the EU AI Act. Carl-Leander Henneking*, *Claas Beger**. Regulatable ML Workshop at NeurIPS 2025, Nov 2025. [Paper](#) 

A Neuroscience-Inspired Dual-Process Model of Compositional Generalization. Alex Noviello*, *Claas Beger**, Jacob Groner, Kevin Ellis, Weinan Sun. NeurIPS 2025 Workshop on Interpreting Cognition in Deep Learning Models, Nov 2025. [Paper](#) 

Towards AI Collaborators: Exploring Goal, Value, and Role-Based Alignment in AI. *Claas Beger*. 51st Annual Meeting of the Society for Philosophy and Psychology (SPP), Jun 2025. [Poster](#)  • **Poster Award**

CoCoNUT: Structural Code Understanding Does Not Fall Out of a Tree. *Claas Beger*, Saikat Dutta. LLM4Code Workshop at ICSE 2025, Jan 2025. [Paper](#) 

Under Review

SpikeDecoder: Realizing the GPT Architecture with Spiking Neural Networks. *Claas Beger*, Florian Walter, Alois Knoll. Under review, Jun 2026. [arXiv](#) 

Citegeist: Automated Generation of Related Work Analysis on the arXiv Corpus. *Claas Beger**, Carl-Leander Henneking*. Under review, Jun 2025. [arXiv](#)  • **Statistics and Data Science Faculty Award**

** indicates equal contribution*

Awards and Recognitions

Diverse Intelligences Fellowship (6000 USD) Awarded by the Natural and Artificial Minds Initiative, Princeton University.	<i>Feb 2026</i>
Poster Award, Society for Philosophy and Psychology (300 USD) Awarded for “Towards AI Collaborators: Exploring Goal, Value, and Role-Based Alignment in AI.”	<i>Jun 2025</i>
Department of Statistics and Data Science Faculty Award (1000 USD) Awarded by Cornell University, for the research project “Citegeist”.	<i>Apr 2025</i>
Dean’s List, Hong Kong University of Science and Technology Top 5% of the undergraduate student body.	<i>Jun 2022</i>